

AN2DL - Second Challenge Report

The Underfitters

Benkirane Ilyas, Lecomte Anatole, Luneau Nathan

295150, 294859, 295387

December 17, 2025

1 Introduction

This report presents the methodology and experimental choices adopted by our team, *The Underfitters*, for the second challenge of the **Artificial Neural Networks and Deep Learning** course.

This challenge addresses an *image classification problem* using **deep learning models** to predict the molecular subtype of diseased human tissue from low-magnification whole-slide images. Unlike the first challenge [1], pretrained architectures and weights are permitted, and models are evaluated using the *F1-score*, encouraging balanced performance across molecular subtypes.

Our approach was inspired by the lectures and practical exercises, particularly the **Lab 7** notebooks on *image augmentation*, *transfer learning*, and *fine-tuning*.

2 Problem Analysis

Dataset characteristics: The task is based on a collection of low-magnification *whole-slide images* of human tissue, accompanied by binary masks highlighting regions likely to contain diseased tissue. A total of 691 of these pairs were used to train the model. The images vary in size, and each sample is labeled with one of four *molecular subtypes*: Luminal A, Luminal B, HER2(+), and Triple Negative.

Main challenges: The main challenges include the limited dataset size, variability in image dimensions, potential class imbalance, and the effective integration of the optional binary masks. During data exploration, we also observed unusual artifacts in some samples. In particular, some images were replaced with unrelated content (ork skin), while others had greenish stain-like regions, potentially harming training if not properly handled.

Initial assumptions: We assumed that data augmentation and transfer learning would be essential given the limited dataset, and that the optional masks could help guide the model. Additionally, we expected it would be necessary to remove images entirely replaced by unrelated content, to ensure reliable training.

3 Method

The methodology consists of two main stages: data management and model experimentation.

3.1 Data Management

The first goal was to establish a robust data processing pipeline that would allow efficient model training and evaluation. The following steps were imple-

mented:

Data exploration: We inspected the dataset, identifying image sizes and the unusual artifacts.

Data preprocessing: All images were resized to 224×224 pixels. Samples fully replaced by artifacts were identified using histogram analysis and removed. For images containing localized greenish stain-like artifacts, the images were kept unchanged, while the corresponding binary masks were modified to remove the blob regions. The images have also been normalized using ImageNet statistics when passing through pretrained models.

Train-validation split: The cleaned dataset was randomly divided into training (70%) and validation (30%) sets, preserving class proportions.

3.2 Model Experimentation

With a working data pipeline in place, the main focus was systematic experimentation with different convolutional neural network architectures and training strategies.

To improve generalization, **data augmentation techniques** such as random flipping, affine transformations, and random erasing were applied to the training set.

Model architectures: We tested several convolutional neural network architectures, including *MobileNetV3Small* (MN), *EfficientNetB0* (EN), and *ResNet18* (RN). For each architecture, three training strategies were considered: *training from scratch*, *transfer learning* with frozen backbone, *fine-tuning* of pretrained weights. All strategies used cross-entropy loss.

Hyperparameter exploration: We experimented with multiple hyperparameters, including learning rate (η), batch size (B), early stopping patience (P), and dropout rate (p_{drop}). Combinations of different models and hyperparameters were initially explored manually, followed by a more systematic grid search to obtain a comprehensive overview of model performance.

Cross-validation: As in the previous project [1], we applied *k-fold cross-validation* to reduce bias from a single train-validation split.

Mask-based input processing: We investigated the use of the binary masks provided with

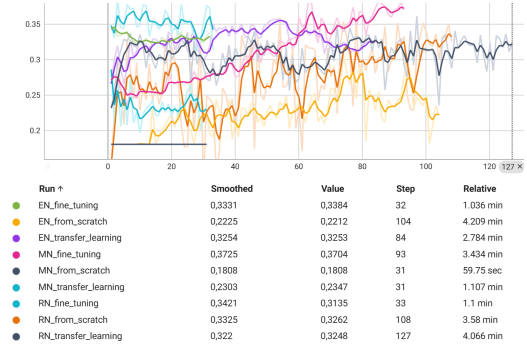
the dataset. For this purpose, each mask was applied directly to its corresponding image, keeping only the masked regions as input to the models.

4 Experiments

Once the experiments were completed on all model architectures introduced in 3.2, we collected their performance metrics on the validation set, including *F1-score* averaged for different random splits (k-fold cross validation), as summarized in Table 1. This allowed us to compare architectures, training strategies, and the impact of mask usage.

Tensorboard visualizations for all models were also generated, showing the evolution of metrics during validation (Figure 1). These visualizations provide insights into convergence, overfitting, and the effect of hyperparameter choices.

Figure 1: Tensorboard visualizations of the experimented CNN models



5 Results

Our best model achieved an F1-score of **32.99%** on the test set. It was obtained using the *EN* architecture with the *transfer learning* training strategy. The hyperparameters of this model were: learning rate $\eta = 10^{-3}$, batch size $B = 64$, dropout rate $p_{\text{drop}} = 0.4$, and early stopping patience $P = 30$.

Despite these promising results, we observed that some architectures failed to converge meaningfully. In particular, certain models showed unstable or flat training curves, indicating that they were either not well-suited to the dataset or were not learning.

Table 1: F1-score of different CNN architectures on the validation set

Strategy	MobileNetV3Small	EfficientNetB0	ResNet18	EfficientNetB0 + Masks
From Scratch	0.1808 $\pm$ 0.0000	0.2342 $\pm$ 0.0432	0.3850 $\pm$ 0.0224	0.2953 $\pm$ 0.0231
Transfer Learning	0.3264 $\pm$ 0.0190	0.3473 $\pm$ 0.0103	0.3624 $\pm$ 0.0120	0.3334 $\pm$ 0.0430
Fine-Tuning	0.3646 $\pm$ 0.0362	0.4049 $\pm$ 0.0254	0.3695 $\pm$ 0.0085	0.3887 $\pm$ 0.0268

6 Discussion

1. Strengths: A strength of our approach is the systematic exploration of multiple convolutional architectures and training strategies. We evaluated *MN*, *EN*, and *RN*, each with training *from scratch*, *transfer learning*, and *fine-tuning*.

Additionally, we tested the effect of applying binary masks to focus on relevant regions, and we applied a variety of data augmentation techniques to improve generalization. This broad experimentation provided valuable insights into the strengths, limitations, and performance impact of different architectures, training strategies, and input preprocessing methods on this dataset.

2. Weaknesses:

Hyperparameter search: Although we performed a grid search to refine hyperparameters, we could not explore all possible combinations of architectures (*MN*, *EN*, and *RN*), training strategies (*from scratch*, *transfer learning*, *fine-tuning*), mask usage, and values of the hyperparameters (η , B , P , and p_{drop}) due to computational constraints. As a result, there likely exists a combination of parameters that would outperform our current best model.

Mask application: Masks were applied directly to the images, effectively removing all background information. While this simplifies input processing, it may discard potentially useful context. Alternative approaches, such as providing the original RGB image along with the mask as an additional channel [2], could potentially yield better results.

Handling of green stains: For images containing greenish stain-like artifacts, we removed the stain from the masks but not from the images. Developing a method to remove or correct these artifacts in the images might have improved model performance.

Data augmentation: Although we applied

standard data augmentation techniques (horizontal flips, affine transformations, random erasing), a more extensive study could be performed to evaluate the impact of different augmentation strategies or larger amounts of augmented images on model performance. Some augmentations may be more effective than others for this dataset, and exploring these options could lead to improvements.

7 Conclusions

In this project, our team The Underfitters focused on classifying diseased human tissue images for the *AN2DL* challenge using deep learning models. Our main contributions include the development of a robust data pipeline with artifact handling, and the use of data augmentation alongside the exploration of several Convolutional Neural Network architectures with different training strategies (from scratch, transfer learning, and fine-tuning) and hyperparameter configurations. These efforts allowed us to achieve an F1-score of 32.99% on the test set.

Improvements and Future Work: Future improvements could address the weaknesses identified in our study. A more extensive, **automated hyperparameter search** is necessary to fully explore the model space defined by the architectures, training strategies, and mask usage.

Future work should investigate **alternative mask integration strategies**, such as using the binary mask as a fourth input channel alongside the image [2], to preserve potentially useful contextual information. Incorporating other data augmentation techniques could further improve the model’s robustness. Finally, a refined method to correct the greenish stain-like artifacts in the images, rather than only in the masks, would ensure more reliable training.

References

- [1] I. Benkirane, A. Lecomte, and N. Luneau. An2dl - first challenge report. Challenge report, The Underfitters, 2025.
- [2] M. Kazemimoghadam, Z. Yang, M. Chen, L. Ma, W. Lu, and X. Gu. Leveraging global binary masks for structure segmentation in medical images. *Physics in Medicine & Biology*, 68(18):185023, 2023.